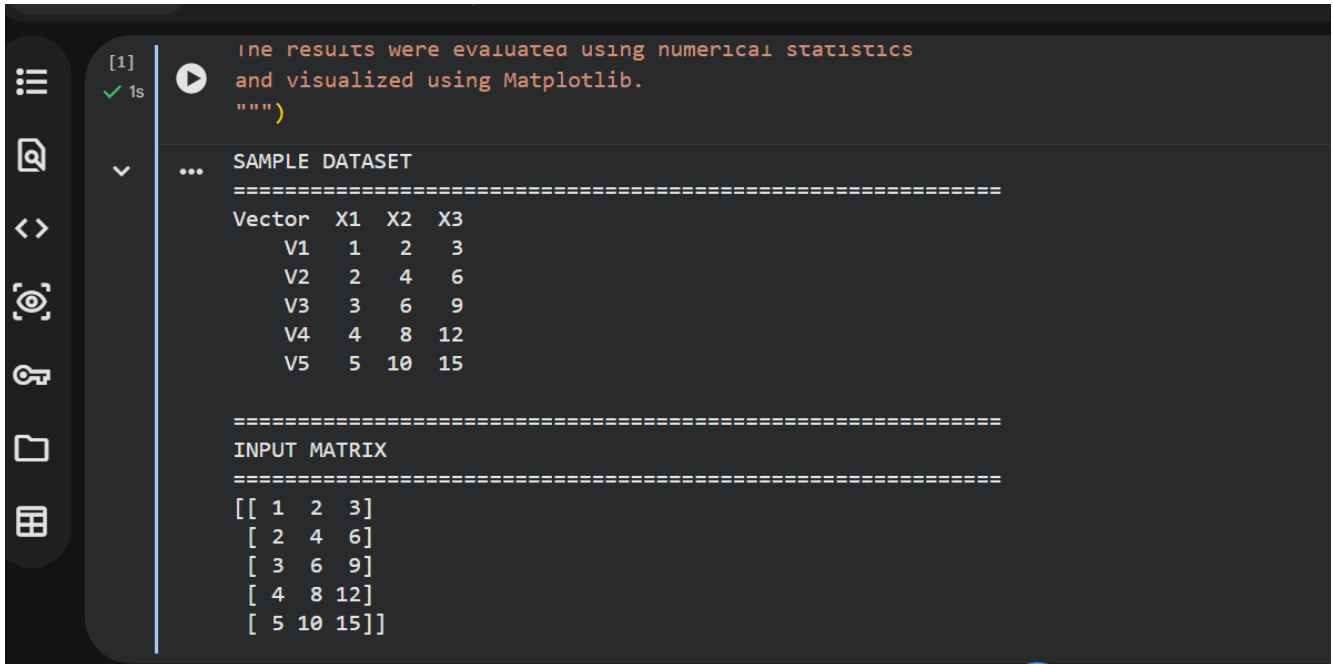


DEEP LEARNING LAB

EXPERIMENT 1

Vector and Matrix Operations for Deep Learning

Practical Output and Result Screenshots



```
[1] ✓ 1s the results were evaluated using numerical statistics
and visualized using Matplotlib.
"""

... SAMPLE DATASET
=====
Vector  X1  X2  X3
V1      1   2   3
V2      2   4   6
V3      3   6   9
V4      4   8  12
V5      5  10  15

=====
INPUT MATRIX
=====
[[ 1  2  3]
 [ 2  4  6]
 [ 3  6  9]
 [ 4  8 12]
 [ 5 10 15]]
```

Figure 1: Sample Dataset and Input Matrix

```
... VECTOR OPERATIONS
=====
Vector V1: [1 2 3]
Vector V2: [2 4 6]
V1 + V2 : [3 6 9]
V2 - V1 : [1 2 3]
V1 · V2 : 28
Magnitude of V1: 3.7417
Magnitude of V2: 7.4833

=====
MATRIX OPERATIONS
=====

Matrix A:
[[1 2]
 [3 4]]
```

Figure 2: Vector Operations

```
[1] ✓ 1s The results were evaluated using numerical statistics
and visualized using Matplotlib.
""" )

... =====
RESULT / EVALUATION TABLE
=====

      Operation  Result
Vector Addition [3 6 9]
Vector Subtraction [1 2 3]
      Dot Product      28
      Magnitude V1  3.7417
      Magnitude V2  7.4833
Matrix Determinant   -2.0

=====
MATRIX STATISTICS
=====
Matrix A Mean: 2.5
Matrix A Maximum: 4
Matrix A Minimum: 1
Matrix A Standard Deviation: 1.118
```

Figure 3: Result / Evaluation Table and Matrix Statistics

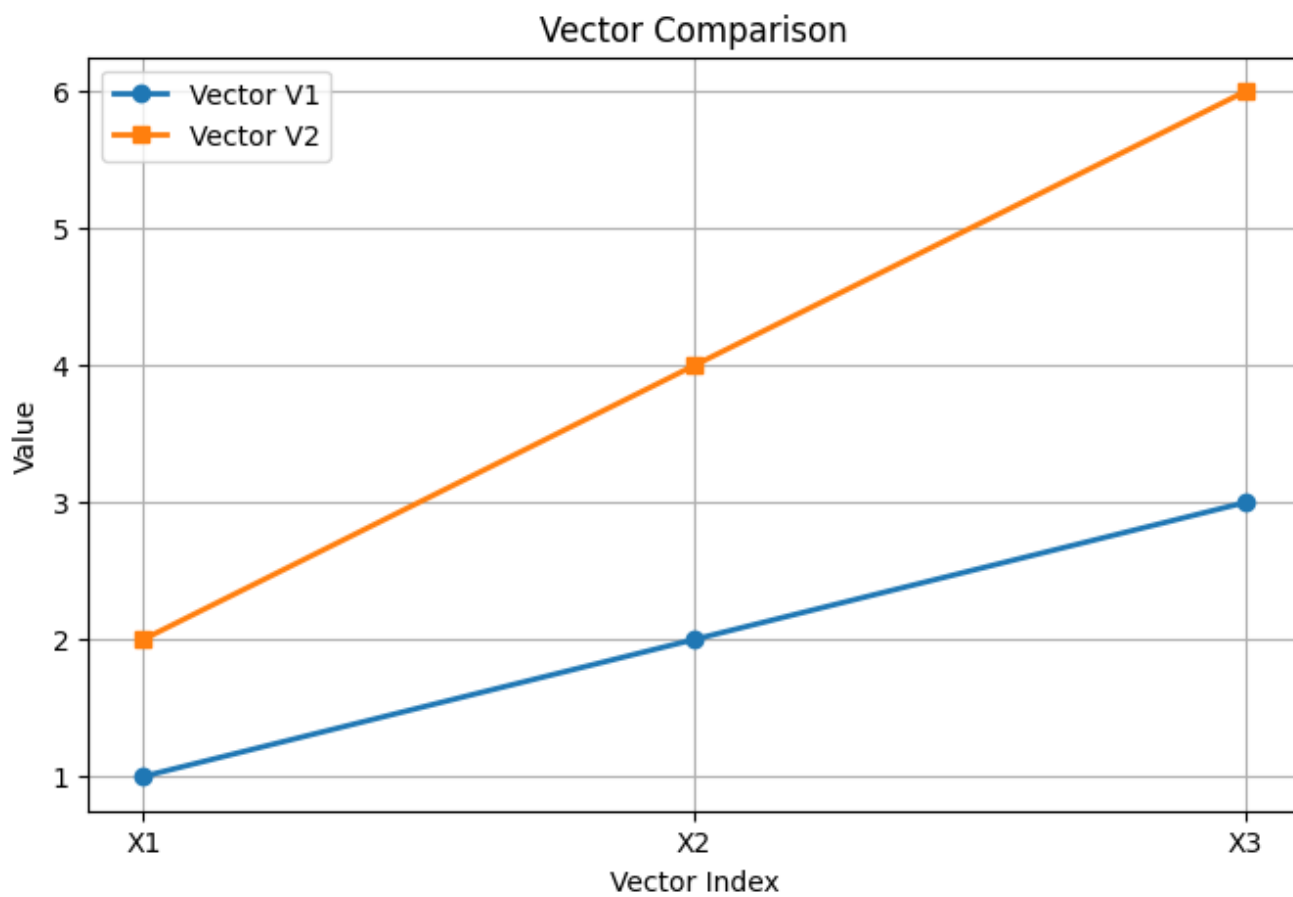


Figure 4: Vector Comparison Plot

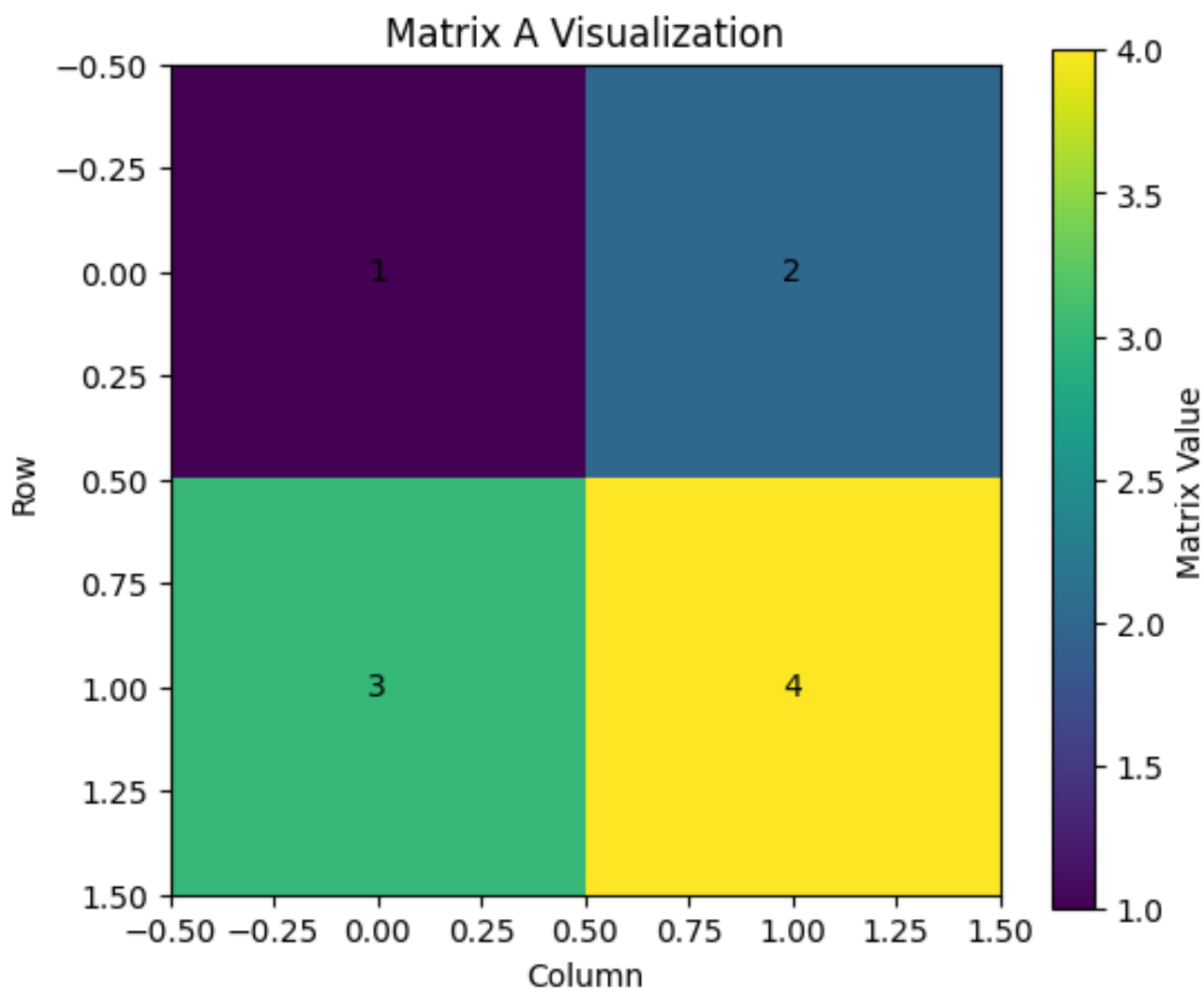


Figure 5: Matrix A Visualization

FINAL RESULT

=====

...

Vector and matrix operations were successfully performed using NumPy.

The experiment demonstrated:

1. Vector addition
2. Vector subtraction
3. Dot product
4. Vector magnitude
5. Matrix addition
6. Matrix subtraction
7. Matrix multiplication
8. Matrix transpose
9. Matrix determinant

The results were evaluated using numerical statistics and visualized using Matplotlib.

Figure 6: Final Result